

Allison Casey

13519 Mira Montana Dr, Del Mar, CA, 92014

allionscasey@gmail.com

858-353-0656

Education

Texas Christian University, Fort Worth, TX

Bachelor of Science in Mechanical Engineering, May 2028

Minor: Innovation and Entrepreneurship

John V. Roach Honors College

Cumulative GPA: 3.87

Experience

R&D Engineering Intern

May 2026-present

Reflow Medical

San Clemente, CA

- Perform medical device builds and dimension inspections, disease research, device bench testing and data analysis
- Create CAD parts and drawings, and powerpoints over data findings
- Write lab notebook studies, testing protocols, work orders, and manufacturing protocols
- Sit in on meetings for physician feedback, manufacturing updates, and cross-divisional checkpoints

Teaching Assistant and Engineering Clinic Staff

August 2025-present

TCU Engineering

Fort Worth, TX

- Support 100+ students weekly by assisting in labs, grading assignments, and running office hours

Private Tutor

January 2025-present

Math Around the Corner

Fort Worth, TX

- Teach middle and high school students in Algebra, Calculus, Physics, and Chemistry

UX Design Team Shadow

July 2023

ServiceNow

San Diego, CA

- Participated in daily stand-up and sprint planning with UX engineers
- Assisted and observed brainstorming sessions for product usability
- Shadowed designers in bug triage and fixes, learned how feedback issues were addressed

Projects, Awards, and Publications

Illuminart - Sensory Lightbox | Rady Children's Hospital Helen Bernardy Center

Designed and built an adaptive, interactive, light box to promote cognitive function for medically fragile children. Integrated an easy-touch switch, brightness control, and ten light modes. Portable, battery-powered, easy to clean.

- Rostam Reifschneider Endowed Award for Leadership, Innovation, and Entrepreneurship
- Best Overall Social Innovation and Photonics Innovation Award - Jacobs Teen Innovation Challenge

The Pop-Shot - Mobile Vaccine Truck Design | Cooper Hewitt National High School Design Competition

Created a mobile vaccine truck to deliver COVID PPF and vaccines to low-income neighborhoods that faced barriers and lacked resources during the initial rollouts of vaccines. Focused on accessibility, efficiency, and equity.

- Top 3 National Finalist
- Published in Ellen Lupton's "[Health Design Thinking: Volume 2.](#)"

Airplane Repair | Texas Christian University Engineering Design Project

Worked in a team of four to design and manufacture an aluminum patch for a damaged aircraft with a missing rivet. Used Finite Element Analysis to test different CAD mockups and optimize the design of the patch for manufacturing. Installed and analyzed strain gauge data to measure stress reduction after patch was applied to the damaged aircraft. Completed a Critical Design Review technical presentation.

Calla - Wearable Tech Bracelet | Texas Christian University Independent Research Project (Incoming, Fall 2026)

Health wearable bracelet designed to blend in like standard jewelry while tracking vital metrics for health and being tailored specifically for women. Focus is on researching how physiological variances in female bodies influence sensor performance and data interpretation.

- Science and Engineering Research Center Grant

Skills

Programming & Simulation: Matlab, Java, Javascript, Arduino (C++), LTspice

CAD & Design: SolidWorks, Autodesk Inventor, Inkscape, Figma

Prototyping & Lab Equipment: Laser Cutter, 3D Printer, Soldering Iron, Power Tools, Oscilloscope, Digital Multimeter, Function Generator

Soft Skills: Public Speaking, Leadership, Time Management, Collaboration, Problem-Solving, Spanish

Relevant Coursework

Engineering: Solid Mechanics I-II, Static Mechanics, Circuit Analysis & Design I-II, Dynamics and Vibrations, Design & Graphics

Programming: Applied Programming-Matlab, Computer Science (honors)

Innovation: Advanced Innovation Technology Studio (Honors), Advanced Design to Entrepreneurship (Honors)

Math & Science: Calculus I-III, Differential Equations, Statistics, Physics I: Mechanics, Physics II: Electromagnetism & Optics, Chemistry

Business: Intro to Microeconomics (Honors), Intro to Accounting, Entrepreneurial Opportunity Recognition, New Venture Development

Soft-Skill Development: Public Address (Honors)

Leadership, Activities, and Memberships

Chi Omega Fraternity | August 2024-present

Society of Women Engineers | August 2024-present

American Society of Mechanical Engineers | January 2025-present

Honors Community Service | January 2025-present

John V. Roach Honors Ambassador | August 2025-present

- Meet with prospective students and give information about the honors college and experience
- Give tours to families and special guests of the honors college and honors dorm
- Attend recruitment events for prospective and transfer students

La Jolla Country Day School Varsity Dance Team Captain | March 2021-May 2024

- Led weekly practices teaching technique, cleaning routines, and building morale
- Choreographed 2 routines per season and managed performances and appearances
- Maintained communication with school administrators and vendors, oversaw purchasing
- Taught 40 K-4th graders and managed team for weekly afterschool dance class each year